

Zekun Wang

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Research Interests

Model/Data-Efficiency & Acceleration: Efficient architecture for Transformers or hybrid ones, Pruning, distillation, quantization .etc to reduce model size and speedup inference, or efficient training LLMs or MLLMs with a small cost (time or data).

Multi-modal Models and Applications: Large multi-modal models (comprehensive, generation, or unify the both), which support diverse tasks and can be applied as agents in digital or embodied environments.

EDUCATION

Ph.D. Student, Harbin Institute of Technology

Major: Computer Science

Harbin, China

Sept. 2019 - present

B.E., Harbin Institute of Technology

Major: Software Engineering

Harbin, China

Sept. 2015 - June 2019

PUBLICATIONS

** denotes equal contributions*

- Aguviz: Unified Pure Vision Agents for Autonomous GUI Interaction.
Yiheng Xu*, **Zekun Wang***, Junli Wang*, Dunjie Lu, Tianbao Xie, Amrita Saha, Doyen Sahoo, Tao Yu, Caiming Xiong.
In: *Forty-Second International Conference on Machine Learning (ICML)*. 2025.
- Improved Diffusion-based Generative Model with Better Adversarial Robustness.
Zekun Wang*, Mingyang Yi*, Shuchen Xue, Zhenguo Li, Ming Liu, Bing Qin, Zhi-Ming Ma.
In: *Thirteenth International Conference on Learning Representations (ICLR)*. 2025.
- AgentTrek: Agent Trajectory Synthesis via Guiding Replay with Web Tutorials.
Yiheng Xu, Dunjie Lu, Zhennan Shen, Junli Wang, **Zekun Wang**, Yuchen Mao, Caiming Xiong, Tao Yu.
In: *Thirteenth International Conference on Learning Representations (ICLR) **Spotlight***. 2025.
- CFSP: An Efficient Structured Pruning Framework for LLMs with Coarse-to-Fine Activation Information.
Yuxin Wang*, Minghua Ma*, **Zekun Wang***, Jingchang Chen, Huiming Fan, Liping Shan, Qing Yang, Dongliang Xu, Ming Liu, Bing Qin.
In: *Proceedings of the 31st International Conference on Computational Linguistics (COLING)*. 2025.
- Demons in the Detail: On Implementing Load Balancing Loss for Training Specialized Mixture-of-Expert Models.
Zihan Qiu, Zeyu Huang, Bo Zheng, Kaiyue Wen, **Zekun Wang**, Rui Men, Ivan Titov, Dayiheng Liu, Jingren Zhou, Junyang Lin.
Preprint. 2025.
- CodeElo: Benchmarking Competition-level Code Generation of LLMs with Human-comparable Elo Ratings.
Shanghaoran Quan, Jiayi Yang, Bowen Yu, Bo Zheng, Dayiheng Liu, An Yang, Xuancheng Ren, Bofei Gao, Yibo Miao, Yunlong Feng, **Zekun Wang**, Jian Yang, Zeyu Cui, Yang Fan, Yichang Zhang, Binyuan Hui, Junyang Lin.
Preprint. 2025.
- Exploring & exploiting high-order graph structure for sparse knowledge graph completion.

- Tao He, Ming Liu, Yixin Cao, **Zekun Wang**, Zihao Zheng, Zheng Chu, Bing Qin.
In *Journal of Frontiers of Computer Science*. 2025.
- Qwen2.5 Technical Report.
Qwen Team.
Technical Report. 2024.
 - CogGPT: Unleashing the Power of Cognitive Dynamics on Large Language Models.
Yaojia Lv, Haojie Pan, **Zekun Wang**, Jiafeng Liang, Yuanxing Liu, Ruiji Fu, Ming Liu, Zhongyuan Wang, Bing Qin.
In: *Findings of the Association for Computational Linguistics: EMNLP*. 2024.
 - Divide-and-Conquer Meets Consensus: Unleashing the Power of Functions in Code Generation.
Jingchang Chen, Hongxuan Tang, Zheng Chu, Qianglong Chen, **Zekun Wang**, Ming Liu, Bing Qin.
In: *Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS) Oral*. 2024.
 - GUIDE: A Guideline-Guided Dataset for Instructional Video Comprehension.
Jiafeng Liang, Shixin Jiang, **Zekun Wang**, Haojie Pan, Zerui Chen, Zheng Chu, Ming Liu, Bing Qin, Ruiji Fu, Zhongyuan Wang.
In: *33rd International Joint Conference on Artificial Intelligence (IJCAI)*. 2024.
 - SmartTrim: Adaptive Tokens and Attention Pruning for Efficient Vision-Language Models.
Zekun Wang^{*}, Jingchang Chen^{*}, Wangchunshu Zhou, Haichao Zhu, Jiafeng Liang, Liping Shan, Ming Liu, Dongliang Xu, Qing Yang, Bing Qin.
In: *2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (COLING-LREC) Oral*. 2024.
 - OFA-Diffusion Compression: Compressing Diffusion Model in One-Shot Manner.
Zekun Wang, Mingyang Yi, Ming Liu, Bing Qin, Zhenguo Li.
In Submission. 2024.
 - MTGER: Multi-view Temporal Graph Enhanced Temporal Reasoning over Time-Involved Document.
Zheng Chu, **Zekun Wang**, Jiafeng Liang, Ming Liu, Bing Qin.
In *Findings of the Association for Computational Linguistics: EMNLP*. 2023.
 - GTR: A Grafting-Then-Reassembling Framework for Dynamic Scene Graph Generation.
Jiafeng Liang, Yuxin Wang, **Zekun Wang**, Ming Liu, Ruiji Fu, Zhongyuan Wang, Bing Qin.
In *32nd International Joint Conference on Artificial Intelligence (IJCAI)*. 2023.
 - TAGNet: A Tiny Answer-Guided Network for Conversational Question Generation.
Zekun Wang, Haichao Zhu, Ming Liu, Bing Qin.
In: *International Journal of Machine Learning and Cybernetics (IJMLC)*. 2023.
 - Distilled Dual-Encoder Model for Vision-Language Understanding.
Zekun Wang, Wenhui Wang, Haichao Zhu, Ming Liu, Bing Qin, Furu Wei.
In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2022.
 - Less Is More: Domain Adaptation with Lottery Ticket for Reading Comprehension
Haichao Zhu, **Zekun Wang**, Heng Zhang, Ming Liu, Sendong Zhao, Bing Qin.
In: *Findings of the Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*. 2021.
 - Molweni: A Challenge Multiparty Dialogues-based Machine Reading Comprehension Dataset with Discourse Structure.
Jiaqi Li, Ming Liu, Min-Yen Kan, Zihao Zheng, **Zekun Wang**, Wenqiang Lei, Ting Liu, Bing Qin.
In: *Proceedings of the 28th International Conference on Computational Linguistics (COLING)*. 2020.

EXPERIENCE

Alibaba Qwen Team.

Research Intern

- Worked on pre-training and model architectures.

Huawei Noah's ark Lab.

Beijing, China

May. 2024 - Present

Beijing, China

Research Intern

Jul. 2023 - Apr. 2024

- Worked on improving the efficiency of diffusion models by model compression and step acceleration.

Microsoft Research Asia (MSRA), Natural Language Computing Group.

Beijing, China

Research Intern

Sept. 2020 - Sept. 2021

- Worked with Researcher Wenhui Wang and Dr. Furu Wei on large-scale pre-trained models and efficient methods (like knowledge distillation) in multimodality and natural language processing.

Joint Laboratory of HIT and iFLYTEK Research, Reading Comprehension Group

Beijing, China

Research Intern

June 2019 - Aug. 2019

- Explored the generalization of pre-trained models on different question answering datasets and got the 3rd place in MRQA 2019 shared task@EMNLP 2019.

PROJECTS

Huozi Chat-LLM

Mar. 2023 - May. 2023

- Participate in instruction turning with LLM and reduce resource consumption of deployment by pruning or quantization.

HONORS & AWARDS

Silver Medal

Shanghai, China

The ACM/ICPC Asia Contest EC-Final

2017

Silver Medal

Shenyang, China

The ACM-ICPC Asia Regional Contest Shenyang Site

2017

Silver Medal

Fuzhou, China

The CCF Collegiate Computer Systems & Programming contest

2017

National Scholarship

Harbin, China

Harbin Institute of Technology

2018

SERVICES

Reviewer: ICML, ICLR, NeurIPS, ACL, EMNLP, NAACL, ACL Rolling Review, COLING, WSDM, AAAI, IJCAI

SKILLS

Programming Languages

Python, C/C++

Tools and Frameworks

PyTorch, FairSeq, Huggingface Transformers, Pytorch Lightning